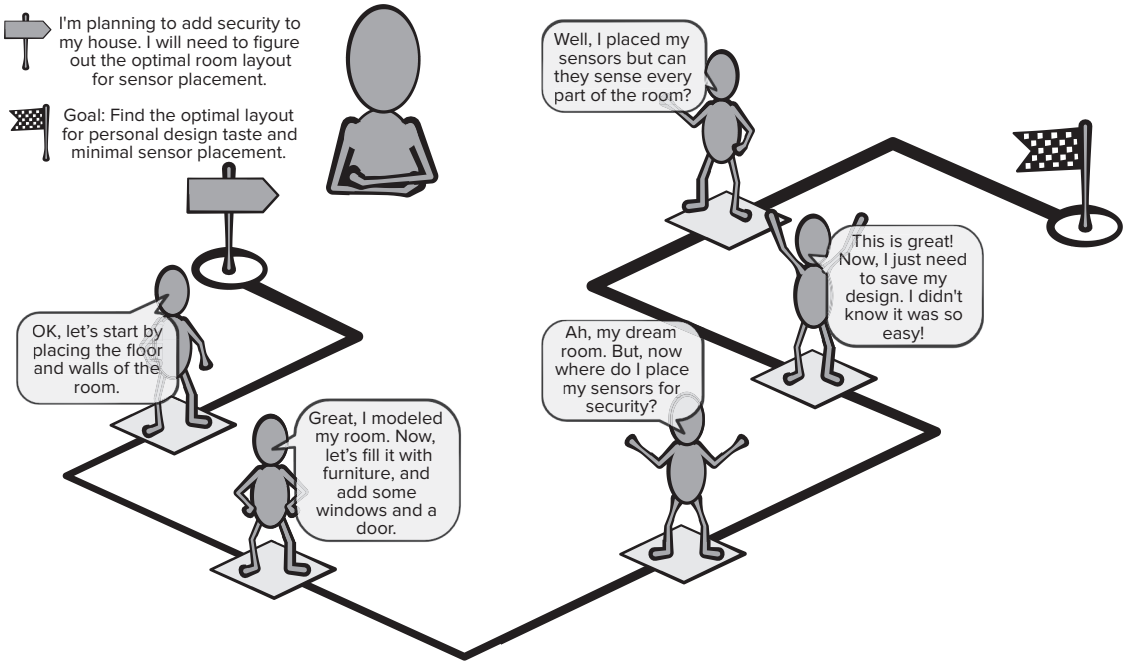
12.4.1 User Research

The phrase *user interface* is probably all the justification needed to spend some time understanding the user before worrying about technical matters. Earlier we noted that each user has a mental image of the software that may be different from the mental image developed by other users. In addition, the user's mental image may be vastly different from the software engineer's design model. The only way that you can get the mental image and the design model to converge is to work to understand the users themselves as well as how these people will use the system. Information from a broad array of sources (user interviews, sales input, marketing input, support input) can be used to accomplish this.

Many UX developers like to create a *customer journey map* (Figure 12.3) as a means of outlining their goals and plans for the software product. The customer journey map shows how the users will experience the software product as if they were traveling on a physical trip with touchpoints (milestones), obstacles, and ways to monitor their progress. Christensen [Chr13] suggests the following steps to create a customer journey map:

1. **Gather stakeholders.** Locate all affected parties needed to ensure diverse viewpoints are included in the customer journey map.
2. **Conduct research.** Collect all information you can about all the things (thoughts, feelings, actions, motivations, expectations, goals, needs,

FIGURE 12.3 Customer journey map



pain points, barriers, questions) users may experience as they use the software product and define your customer phases. The customer phases will become your touchpoints and are shown as labeled squares in Figure 12.3.

3. **Build the model.** Create a visualization of the touch points (any interaction between the user and product), channels (interaction devices or information streams), and actions taken by the customer (user).
4. **Refine the design.** Recruit a designer to make the deliverable visually appealing and ensure that customer phases are identified clearly.
5. **Identify gaps.** Note any gaps in the customer experience or points of friction or pain (places with information overlap or poor transition between phases).
6. **Implement your findings.** Assign responsible parties to bridge the gaps and resolve pain points found.

12.4.2 User Modeling

In previous chapters, you learned that the user story describes the manner in which an actor (in the context of user interface design, an actor is always a person) interacts with a system. When used as part of task analysis, the user story is refined to become a formal use case to show how an end user performs some specific work-related task. In most instances, the use case is written in an informal style (a simple paragraph) in the first person. For example, assume that a small software company wants to build a computer-aided design system explicitly for interior designers. To get a better understanding of how they do their work, actual interior designers are asked to describe a specific design function. When asked, "How do you decide where to put furniture in a room?" an interior designer writes the following informal user story:

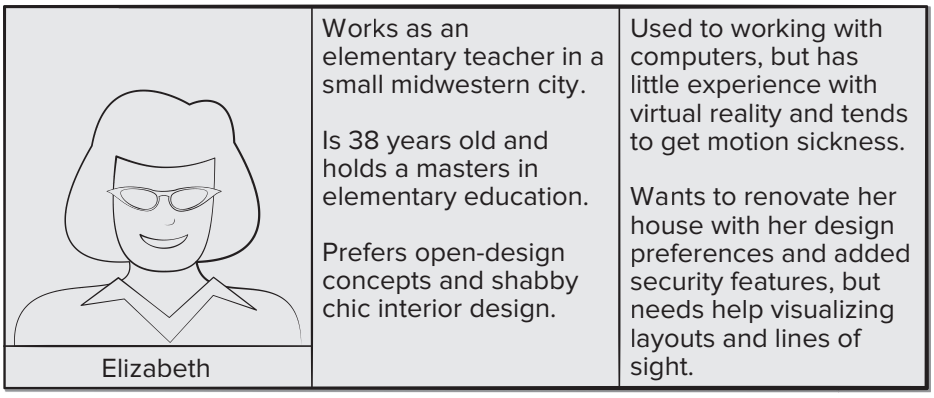
> I begin by sketching the floor plan of the room, the dimensions and the location of windows and doors. I'm very concerned about light as it enters the room, about the view out of the windows (if it's beautiful, I want to draw attention to it), about the running length of an unobstructed wall, about the flow of movement through the room. I then look at the list of furniture my customer and I have chosen. . . . Then, I draw a rendering (a 3-D picture) of the room to give my customer a feel for what it'll look like.

This user story provides a basic description of one important work task for the computer-aided design system from the perspective of one of its users. From it, you can extract tasks, objects, and the overall flow of the interaction. In addition, other features of the system that would please the interior designer might also be conceived. For example, a digital photo could be taken looking out each window in a room. When the room is rendered, the actual outside view could be represented through each window. However, if there is more than one type of user, it may be important to define more than one set of user goals for the system described by the user stories.

User experience designers often create fictional user personas to summarize the assumptions made for the different types of users. A *user persona* is a representation of the goals and behavior of a hypothesized group of users. Personas are often

FIGURE 12.4

Persona example



synthesized from data collected during interviews with users. Figure 12.4 shows an example of a user persona. Personas are often used to improve product designers' abilities to see through the eyes of target users [Hil17].

Lene Nielsen [Nie13] describes four tasks that occur during the general process of creating and using personas to guide the user experience design process:

- **Data collection and analysis.** Stakeholders collect as much information about the proposed product users as possible to determine the user groups and begin to emphasize what each group needs.
- **Describe personas.** The developers need to decide how many personas it is reasonable to create, if they will create more than one, and which persona will be their primary focus. The developers create and name each persona including details about their education, lifestyle, values, goals, needs, limitations, desires, attitudes, and behavior patterns.
- **Develop scenarios.** Scenarios are user stories about how personas will use the product being developed. They may focus on the touchpoints and obstacles described in the customer journey. They should show how personas would overcome problems using the system resources if they were the actual users of the system.
- **Acceptance by stakeholders.** Often this is done by validating the scenarios using a review technique or demonstration called *cognitive walkthrough*.[3] Stakeholders assume the role defined by the persona and work through a scenario using a system prototype.

3 A cognitive walkthrough is a method that evaluates whether the order of cues and prompts in a system supports the way people process tasks and anticipates the "next steps" of a system by having users verbalize their decision-making process while using a system representation to complete a user goal.

SAFEHOME



Use Cases for User Interface Design

The scene: Vinod's cubicle, as user interface design continues.

The players: Vinod and Jamie, members of the *SafeHome* software engineering team.

The conversation:

Jamie: I pinned down our marketing contact and had her write a user story for the surveillance interface.

Vinod: From whose point of view?

Jamie: The homeowner, who else is there?

Vinod: There's also the system administrator role. Even if it's the homeowner playing the role, it's a different point of view. The administrator sets the system up, configures stuff, lays out the floor plan, places the cameras . . .

Jamie: All I had her do was play the role of the homeowner when he wants to see video.

Vinod: That's okay. It's one of the major behaviors of the surveillance function interface. But we're going to have to examine the system administrator's behavior as well.

Jamie (irritated): You're right.

(Jamie leaves to find the marketing person. She returns a few hours later.)

Jamie: I was lucky. I found her, and we worked through the administrator's user story together using the homeowner persona. Basically, we're going to define "administration" as one function that's applicable to all other *SafeHome* functions. Here's what we came up with.

(Jamie shows the user story to Vinod.)

User story: I want to be able to set up or edit the system layout at any time. When I set up the system, I select an administration function. It asks me whether I want to do a new setup or whether I want to edit an existing setup. If I select a new setup, the system displays a drawing screen that will enable me to draw the floor plan onto a grid. There will be icons for walls, windows, and doors so that drawing is easy. I just stretch the icons to their appropriate lengths. The system will display the lengths in feet or meters (I can select the measurement system). I can select from a library of sensors and cameras and place them on the floor plan. I get to label each, or the system will do automatic labeling. I can establish settings for sensors and cameras from appropriate menus. If I select edit, I can move sensors or cameras, add new ones or delete existing ones, edit the floor plan, and edit the setting for cameras and sensors. In every case, I expect the system to do consistency checking and to help me avoid mistakes.

Vinod (after reading the scenario): Okay, there are probably some useful design patterns [Chapter 14] or reusable components for GUIs for drawing programs. I'll betcha 50 bucks we can implement some or most of the administrator interface using them.

Jamie: Agreed. I'll check it out.

12.4.3 Task Analysis

The user's goal is to accomplish one or more tasks via the software product. To accomplish this, the software user interface must provide mechanisms that allow the user to achieve her goal. The goal of task or scenario analysis is to answer the following questions:

- What work will the user perform in specific circumstances?
- What tasks and subtasks will be performed as the user does the work?

- What specific problem domain objects will the user manipulate as work is performed?
- What is the sequence of work tasks—the workflow?
- What is the hierarchy of tasks?

To answer these questions, you must draw upon techniques that we have discussed earlier in this book, but in this instance, these techniques are applied to the user interface.

In Chapter 8, we discussed stepwise elaboration (also called functional decomposition or stepwise refinement) as a mechanism for refining the processing tasks that are required for software to accomplish some desired function. Task analysis for interface design uses an elaborative approach that assists in understanding the human activities the user interface must accommodate.

First, you should define and classify the human tasks that are required to accomplish the goal of the system or app. For example, let's reconsider the computer-aided design system for interior designers discussed earlier. By observing an interior designer at work, you notice that interior design comprises a number of major activities: furniture layout (note the user story discussed earlier), fabric and material selection, wall and window coverings selection, presentation (to the customer), costing, and shopping. Each of these major tasks can be elaborated into subtasks. For example, using information contained in the use case, furniture layout can be refined into the following tasks: (1) draw a floor plan based on room dimensions, (2) place windows and doors at appropriate locations, (3a) use furniture templates to draw scaled furniture outlines on the floor plan, (3b) use accents templates to draw scaled accents on the floor plan, (4) move furniture outlines and accent outlines to get the best placement, (5) label all furniture and accent outlines, (6) draw dimensions to show location, and (7) draw a perspective-rendering view for the customer. A similar approach could be used for each of the other major tasks.

Subtasks 1 to 7 can each be refined further. Subtasks 1 to 6 will be performed by manipulating information and performing actions within the user interface. On the other hand, subtask 7 can be performed automatically in software and will result in little direct user interaction.[4] The design model of the interface should accommodate each of these tasks in a way that is consistent with the user model (the profile of a "typical" interior designer) and system perception (what the interior designer expects from an automated system).

12.4.4 Work Environment Analysis

Hackos and Redish [Hac98] discuss work environment analysis this way: "People do not perform their work in isolation. They are influenced by the activity around them, the physical characteristics of the workplace, the type of equipment they are using, and the work relationships they have with other people." In some applications the user interface for a computer-based system is placed in a "user-friendly location"

4 However, this may not be the case. The interior designer might want to specify the perspective to be drawn, the scaling, the use of color, and other information. The use case related to drawing perspective renderings would provide the information you need to address this task.

(e.g., proper lighting, good display height, easy keyboard access), but in others (e.g., a factory floor or an airplane cockpit), lighting may be suboptimal, noise may be a factor, a keyboard or mouse or touch screen may not be an option, or display placement may be less than ideal. The interface designer may be constrained by factors that mitigate against ease of use.

In addition to physical environmental factors, the workplace culture also comes into play. Will system interaction be measured in some manner (e.g., time per transaction or accuracy of a transaction)? Will two or more people have to share information before an input can be provided? How will support be provided to users of the system? These and many related questions should be answered before the interface design commences.

12.5 User Experience Design

As with all iterative processes, there is not a clean division between analysis and design. Figure 12.5 illustrates the practice of cycling between user research and design, as well as cycling between design and construction. The emphasis is on creating incremental prototypes and by testing it with actual users.

Google has defined a 5-day sprint for doing UX design [Kna16], [Goo18], [DXL18]. The steps are outlined below with one day allocated to each:

- **Understand.** Encompasses the user research activities in which the team gathers the information on the problem to be solved (user needs and business goals) for the software product; one way of doing this is to have a series of

FIGURE 12.5
Iterative UX design process

